

What Causes Chronic Migraine?

A migraine is a severe type of headache that causes throbbing pain on one side of your head. Nerve fibers in your brain blood vessels activate, causing pain. Those nerve fibers move through the meninges, which is the protective membrane around your brain and spinal cord.¹

Some of the most common triggers of migraines include:¹

- Anxiety and stress
- Bright, flashing lights
- Depression
- Exerting yourself too hard
- Loud noises
- Low blood sugar and skipped meals
- Motion sickness
- Poor sleep quality
- Strong fumes and odors
- Sudden weather or environmental changes

How Chronic Migraine Is Diagnosed

A healthcare provider can diagnose chronic migraine based on your symptoms and health history. They might perform a physical exam and order diagnostic tests to see if a brain disorder, muscle tension, or sinus problems cause your migraines.²

Diagnostic tests include:²

- **Brain CT or head MRI:** These imaging tests take detailed images of your brain. A healthcare provider might use an imaging test if you have a loss of alertness, memory problems, and weakness.
- **Electroencephalogram (EEG):** This test measures electrical activity in your brain to rule out seizures.³
- **Lumbar puncture (spinal tap):** A healthcare provider will collect a sample of cerebrospinal fluid (CSF), which cushions your brain and spinal cord. A spinal tap rules out neurologic disorders, including meningitis (an infection of the meninges).⁴

How To Prevent Chronic Migraine

A healthcare provider might advise or prescribe preventative medications if you have migraines more than twice per week. Those medications help reduce how often and how severe your migraines are.²

Other ways to prevent migraines include:¹

- Avoiding foods and drinks that trigger attacks
- Consistently eating meals
- Getting plenty of sleep each night
- Maintaining healthy body weight
- Regularly exercises
- Staying hydrated
- Stopping medications that trigger attacks if possible

Treatment

Migraine treatment is aimed at stopping symptoms and preventing future attacks.

Medicines

Many medicines have been designed to treat migraines. They fall into two broad categories:

- **Pain-relieving medicines.** Also known as acute or abortive treatments, these medicines are taken during migraine attacks and are designed to stop symptoms.
- **Preventive medicines.** These medicines are taken regularly, often daily, to reduce migraine symptoms.

Your treatment choices depend on how often you have headaches and how painful they are. Your treatment also depends on whether you have nausea and vomiting, whether your headaches are disabling, and whether you have other medical conditions.

Pain-relieving medicines

Medicines used to relieve migraine pain work best when taken as soon as symptoms begin. These medicines include:

- **Pain relievers.** Pain relievers that you can get with or without a prescription include aspirin and ibuprofen (Advil, Motrin IB, others). Medicines that combine caffeine, aspirin and acetaminophen (Excedrin Migraine) may be helpful, but usually only against mild migraine pain. But be sure to talk with your healthcare professional about how often it's safe to take these medicines. When taken too

long, they can cause medication overuse headaches and possibly ulcers and bleeding in the gastrointestinal tract.

- **Triptans.** These prescription medicines can relieve many symptoms of migraine by blocking pain pathways in the brain. Triptans are taken as pills, shots, nasal sprays, nasal powders and disintegrating tablets. They include sumatriptan (Imitrex, Tosymra, others), rizatriptan (Maxalt), zolmitriptan (Zomig), naratriptan, almotriptan, eletriptan (Relpax) and frovatriptan (Frova). They might not be safe to take if you're at risk of a stroke or heart attack.
- **Dihydroergotamine (Migranal, Trudhesa).** Available as a nasal spray, this medicine is most effective when taken shortly after the start of symptoms for migraines that tend to last longer than 24 hours. Side effects can include worsening of migraine-related vomiting and nausea. People with coronary artery disease, high blood pressure, or kidney or liver disease should avoid dihydroergotamine.
- **Lasmiditan (Reyvow).** This is an oral medicine approved for the treatment of migraines with or without aura. In studies, lasmiditan significantly improved headache pain. Lasmiditan can make you sleepy and cause dizziness, so don't drive or operate machinery for at least eight hours after taking it.
- **Oral calcitonin gene-related peptides antagonists, also known as gepants.** Ubrogepant (Ubrovelvy) and rimegepant (Nurtec ODT) are oral gepants approved for the treatment of migraines in adults. In clinical trials, medicines from this class were more effective than placebo at relieving pain two hours after taking them. They also were effective at treating migraine symptoms such as nausea and sensitivity to light and sound. Common side effects include dry mouth, nausea and too much sleepiness. Ubrogepant and rimegepant should not be taken with strong CYP3A4 inhibitor medicines such as some medicines used to treat cancer.
- **Intranasal zavegepant (Zavzpret).** This nasal spray is a gepant that can relieve migraine pain within two hours and for up to 48 hours. It also can improve other symptoms, such as nausea and sensitivity to light and sound. Side effects of zavegepant include a change in your sense of taste, nasal discomfort and throat irritation.
- **Antinausea medicines.** These are usually taken with pain medicines and can help if you have nausea and vomiting with your migraines. Medicines include chlorpromazine, metoclopramide (Gimoti, Reglan) and prochlorperazine (Compro).

- **Opioid and barbiturate medicines.** Opioids or barbiturates may help if you can't take other migraine medicines. However, they're usually avoided due to better and more effective options. Because they can be highly addictive, these medicines are used only if no other treatments are effective.

Some migraine medicines are not safe to take during pregnancy. If you're pregnant or trying to get pregnant, don't use any of these medicines without first talking with your healthcare professional.

Neuromodulation therapy

Devices that stimulate nerves, known as neuromodulation devices, can reduce the number of migraines you have or make them less painful. The devices are noninvasive and work by stimulating a peripheral or cranial nerve to change the brain's response to pain. Therapies include transcutaneous supraorbital nerve stimulation, external vagal nerve stimulation and distal transcutaneous electrical stimulation. Combined occipital and trigeminal neurostimulation is another option.

Another therapy called single-pulse transcranial magnetic stimulation treats the brain to relieve migraine pain. For this therapy, you place the device on the back of your head.

Lifestyle and home remedies

When symptoms of migraine start, try heading to a quiet, darkened room. Close your eyes and rest or take a nap. Place a cool cloth or ice pack wrapped in a towel or cloth on your forehead and drink lots of water.

To help soothe migraine pain, you also might want to:

- **Try relaxation techniques.** Breathing exercises and other forms of relaxation training teach you ways to deal with stressful situations, which might help reduce the number of migraines you have.
- **Develop a sleeping and eating routine.** Don't sleep too much or too little. Set and follow a consistent sleep and wake schedule daily. Try to eat meals at the same time every day.
- **Drink plenty of fluids.** Staying hydrated, particularly with water, might help.
- **Keep a headache diary.** Recording your symptoms in a headache diary can help you learn more about what triggers your migraines and what treatment is most effective. It also can help your healthcare professional diagnose your condition and track your progress between visits.
- **Exercise regularly.** Regular aerobic exercise reduces tension and can help prevent a migraine. If your healthcare professional agrees, choose aerobic

activities you enjoy, such as walking, swimming and cycling. Warm up slowly, however, because sudden, intense exercise can cause headaches.

Regular exercise also can help you lose weight or maintain a healthy body weight. Obesity is thought to be a factor in migraines.

Migraines: Simple steps to head off the pain

Migraines cause pain as real as the pain of injuries — with one difference: Healthy habits and simple remedies sometimes stop migraines before they start. [By Mayo Clinic Staff](#)

Medicine is a proven way to both treat and prevent migraines. But medicine is only part of the story. It's also important to take good care of yourself and understand how to cope with migraine pain when it strikes.

The same lifestyle choices that promote overall good health also can reduce the number of migraines you have and lessen the migraine pain.

Combining medicine with behavioral measures and lifestyle can often be the most effective way to handle migraines.

Find a calm environment

At the first sign of a migraine, take a break and step away from whatever you're doing if possible.

- **Turn off the lights.** Light and sound can make migraine pain worse. Relax in a dark, quiet room. Sleep if you can.
- **Try temperature therapy.** Apply hot or cold compresses to your head or neck. Ice packs have a numbing effect, which may dull the pain. Hot packs and heating pads can relax tense muscles. Warm showers or baths may have a similar effect.
- **Sip a caffeinated drink.** In small amounts, caffeine alone can relieve migraine pain in the early stages. Caffeine also may enhance the pain-reducing effects of acetaminophen (Tylenol, others) and aspirin.

Be careful, however. Drinking too much caffeine too often can lead to withdrawal headaches later on. And having caffeine too late in the day may interfere with sleep, which can affect migraines.

Sleep well

Migraines may keep you from falling asleep or wake you up at night. Likewise, migraines are often triggered by a poor night's sleep.

Here are some tips to encourage sound sleep.

- **Establish regular sleep hours.** Wake up and go to bed at the same time every day — even on weekends. If you nap during the day, keep it short. Naps longer than 20 to 30 minutes may interfere with nighttime sleep.
- **Unwind at the end of the day.** Anything that helps you relax can promote better sleep: Listen to soothing music, soak in a warm bath or read a favorite book.

But watch what you eat and drink before bedtime. Intense exercise, heavy meals, caffeine, nicotine and alcohol can interfere with sleep.

- **Lessen distractions.** Save your bedroom for sleep and intimacy. Don't watch television or take work materials to bed. Close your bedroom door. Use a fan to muffle distracting noises.
- **Don't try so hard to sleep.** The harder you try to sleep, the more awake you'll feel. If you can't fall asleep, read or do another quiet activity until you become drowsy.
- **Check your medicine.** Medicine that contains caffeine or another stimulant — including some medicine to treat migraines — may interfere with sleep.

Eat wisely

Your eating habits can influence your migraines. Consider the basics:

- **Be consistent.** Eat at about the same time every day.
- **Don't skip meals.** Fasting increases the risk of migraines.
- **Keep a food journal.** Keeping track of the foods you eat and when you have migraines can help you find potential food triggers.
- **Avoid foods that trigger migraines.** If you suspect that a certain food is triggering migraines, remove it from your diet to see what happens. These foods may include aged cheese, chocolate, caffeine and alcohol.

Exercise regularly

During physical activity, your body releases certain chemicals that block pain signals to your brain. These chemicals also help reduce anxiety and depression — two conditions that can make migraines worse.

Obesity also increases the risk of chronic headaches. Maintaining a healthy weight through exercise and diet can provide additional benefits in managing migraines.

If your health care provider agrees, choose any exercise you enjoy. Walking, swimming and cycling are often good choices. Just remember to ease into exercise gradually, as very vigorous exercise may trigger migraines.

Manage stress

Stress and migraines often go hand in hand. You can't avoid daily stress, but you can keep it under control to help manage your migraines:

- **Simplify your life.** Don't look for ways to squeeze more activities or chores into the day. Instead, find a way to leave some things out.
- **Manage your time wisely.** Update your to-do list every day — both at work and at home. Delegate what you can, and divide large projects into manageable chunks.
- **Take a break.** If you feel overwhelmed, a few slow stretches or a quick walk may renew your energy for the task at hand.
- **Adjust your attitude.** Stay positive. If you find yourself thinking, "This can't be done," switch gears. Think instead, "This will be tough. But I can make it work."
- **Enjoy yourself.** Find time to do something you enjoy for at least 15 minutes every day. It could be playing a game, having coffee with a friend or pursuing a hobby. Doing something you enjoy is a natural way to combat stress.
- **Relax.** Deep breathing from your diaphragm can help you relax. Focus on inhaling and exhaling slowly and deeply for at least 10 minutes every day. It may help to relax your muscles, one group at a time. When you're done, sit quietly for a minute or two.

Keep a migraine diary

A diary may help you determine what triggers your migraines. Note when your migraines started, what you were doing at the time, how long they lasted and what, if anything, provided relief.

Until recently, avoiding migraine triggers was considered the best advice. But new research suggests this may actually increase sensitivity to potential triggers.

A more useful approach may be to gradually expose yourself to triggers, and learn to cope with these headache triggers by using behavioral management techniques. These may include identifying and challenging negative thoughts, relaxation training, and stress reduction. More research is needed to understand if and how this approach is more effective in managing migraines.

Strive for balance

Living with migraines is a daily challenge. But making healthy lifestyle choices can help. Ask your friends and loved ones for support.

If you're feeling anxious or depressed, consider joining a support group or seeking counseling. Believe in your ability to take control of the pain.

Alternative medicine

Nontraditional therapies might help with chronic migraine pain.

- **Acupuncture.** Clinical trials have found that acupuncture may be helpful for headache pain. In this treatment, a practitioner inserts many thin, disposable needles into several areas of your skin at defined points.
- **Biofeedback.** Biofeedback appears to be effective in relieving migraine pain. This relaxation technique uses special equipment to teach you how to monitor and control certain physical responses related to stress, such as muscle tension.
- **Cognitive behavioral therapy.** Cognitive behavioral therapy may benefit some people with migraines. This type of psychotherapy teaches you how behaviors and thoughts affect how you perceive pain.
- **Meditation and yoga.** Meditation may relieve stress, which is a known trigger of migraines. Done on a regular basis, yoga may reduce the number of migraines you have or how long they last.
- **Herbs, vitamins and minerals.** There is some evidence that the herbs feverfew and butterbur might prevent migraines or reduce their severity, though study results are mixed. Butterbur isn't recommended because of safety concerns.

A high dose of riboflavin, also called vitamin B-2, may reduce the number of migraines you have and how painful they are. Coenzyme Q10 supplements might decrease the frequency of migraines, but larger studies are needed.

Magnesium supplements have been used to treat migraines, but with mixed results.

Ask your healthcare professional if these treatments are right for you. If you're pregnant, talk with your healthcare professional before trying any of these treatments.

A Quick Review

Migraine pain is typically something that you cannot ignore. There are preventative, acute, and complementary treatments available, depending on the severity and frequency of your migraines. Those medications may not be right for every person. Talk to a healthcare provider about how to best treat your migraine symptoms.